

The variable name to be used for the `API_TOKEN` is `token`.

Sample request

For example, If we wish to push the value of 29.5 as the temperature value for device `D1`, we will need to do a HTTP POST, as follows: `https://things.ubidots.com/api/v1.6/devices/D1/temperature/values/?token=API_TOKEN`

...with the POST body as follows:

```
{ "value": "29.5" }
```

We can also provide the context and timestamp variables that are optional but can be useful for tracking the location of the device, and also the timestamp at which the sensor value was recorded. The data that will be passed in the HTTP POST request body will then be as follows:

```
{ "value": "29.5", "timestamp": "1464661369000", "context": { "lat": "19.07", "lon": "72.87" } }
```

When you send the context, the Ubidots dashboard that we saw earlier will recognise it as a GeoPoint and will plot it on the map.

Sample transactions

We can use the curl utility to do a few sample HTTP POST requests to our endpoint, so that we can see how we can then visualise the data in the Ubidots dashboard.

Here is a sample curl request for inserting temperature sensor data read at device `D1`:

```
$ curl -X POST -H "Content-Type: application/json" -d '{"value":29.7, "context": {"lat":19.07, "lng":72.87}, "timestamp":1493817120}' http://things.ubidots.com/api/v1.6/devices/D1/temperature/values?token=u31S5y81GakrD64Dw0JsKwYEcje9nv
```

```
{ "url": "http://things.ubidots.com/api/v1.6/values/5909899e7625421b68f86883", "value": 29.7, "timestamp": 1493817120 }
```

```
1493817120, "context": { "lat": 19.07, "lng": 72.87 }, "created_at": "2017-05-03T07:41:18.226" }
```

Similarly, create a few more values with different timestamps and temperature values to simulate data being received from the sensor.

If you visit the Web portal and click on *Devices*, you will find the value has got recorded, as shown in Figure 6.

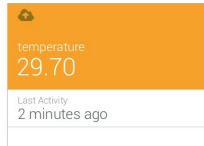


Figure 6: Sample temperature value

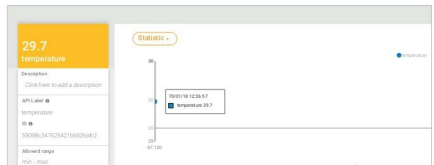


Figure 7: Temperature data

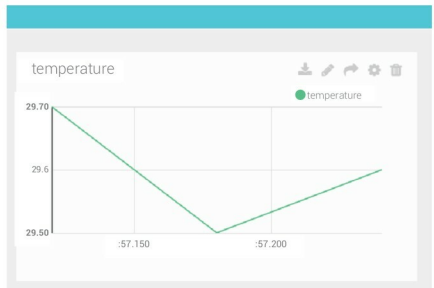


Figure 8: Temperature dashboard

You can even click on the variable, i.e., temperature, to see the details in the graph (Figure 7).

Dashboard

The Ubidots dashboard allows you to create dashboards where you can add graphical widgets to visualise your sensor data from various devices. To create a dashboard, you can go to the Web portal and then click on *Dashboard*. Simply add a new dashboard, select a chart type from one of the many options offered (for example LineChart), and then select the device and the variable. In our case, it is device `D1` and the variable *Temperature*.

Once you have added that to the dashboard, you can see it as shown in Figure 8.